8x32 LED Matrix Frame with Diffuser Grid

fractalnoise

VIEW IN BROWSER

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Summary

No glue, tape or screws needed. Snap-fit sockets for ESP32, mic & power connector. Wall- or desk-mounted.

Gadgets > Other Gadgets

Tags: led esp32 wled ledmatrix ledmatrixcase 8x32ledmatrixcase 8x32ledmatrix

8x32 LED Matrix Frame

- No glue*, tape or screws needed
- Accessible USB port
- Wall- or desk-mounted
- Snap-fit sockets for ESP-WROOM-32, MAX4466 or MAX9814 microphone and DC Power Connector for easy installation
- Optional socket-less version

I suggest gluing the two front frame pieces together

Instructions

Print:

- 2xFront SPLIT
- 2xDiffuser SPLIT
- 1xBackL SPLIT
- 1xBackR SPLIT

Print the Diffuser Screens with these settings to give the screen a horizontal “scanline” texture:

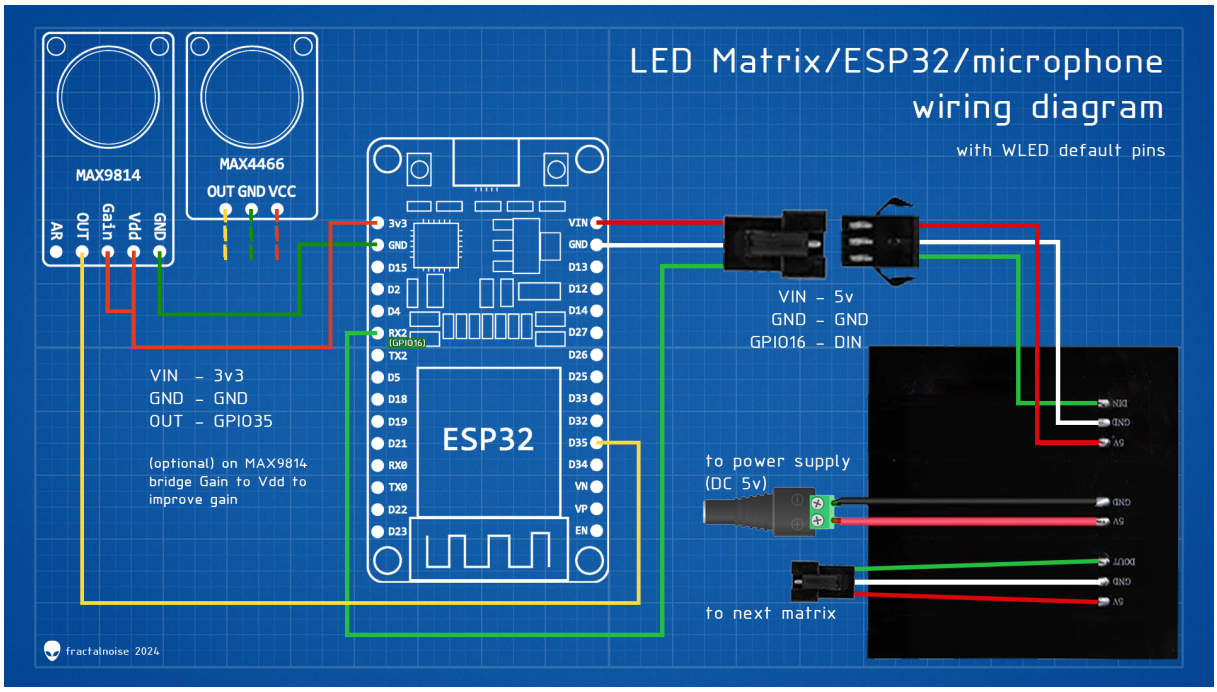
- Top/Bottom surface pattern: ALIGNED RECTILINEAR
- Internal solid infill pattern: ALIGNED RECTILINEAR
- Sparse infill pattern: ALIGNED RECTILINEAR
- Infill direction: 0

If using white filament for the Diffuser Screen, printing it on a textured plate will make it scatter more light and be dimmer. Use a smooth plate for best results. A textured plate will work great for clear filament though :)

For the LED matrix, you'll need:

- 8x32 LED Matrix - e.g. https://aliexpress.com/item/4000544411175.html
- ESP-Wroom-32 - e.g. https://aliexpress.com/item/1005005972312714.html
- MAX4466 microphone - e.g. https://aliexpress.com/item/3256803608057399.html
  or
  MAX9814 microphone - e.g. https://aliexpress.com/item/3256805923391884.html
- Power Supply 5V 1.5A-15A (~2-5A will do for most uses) - e.g. https://aliexpress.com/item/32844702891.html or https://aliexpress.com/item/1005006126703620.html
  or an old USB cable (cut one end and connect the + and - to the LED matrix)

LED Matrix/ESP32/WLED Assembly Guide:



Some useful resources:

- WLED Project: https://kno.wled.ge/
- Chris Maher WLED tutorials playlist: https://www.youtube.com/playlist?list=PLur8dX5PE-4pk838s_gZAdm-98-sf0f0Z
- Make It Work WLED tutorials playlist: https://www.youtube.com/playlist?list=PLUTKOGJvdIQ8WYfG6TDJ_c8_WlPovWbL9

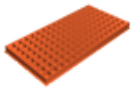
My other LED projects:

https://www.printables.com/@fractalnoise/collections/1236469

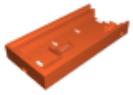
Model files

SPLIT - 3mf 5 files

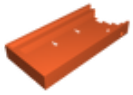
8x32-led-matrix-frame-front-split.3mf



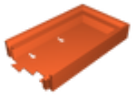
8x32-led-matrix-frame-diffuser-split.3mf



8x32-led-matrix-frame-backl-split.3mf



8x32-led-matrix-frame-backl-no-sockets-split.3mf

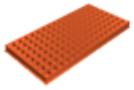


8x32-led-matrix-frame-backr-split.3mf

SPLIT - stl

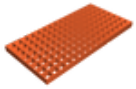
7 files

8x32-led-matrix-frame-front-split.stl

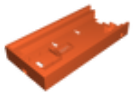


8x32-led-matrix-frame-diffuser-split.stl

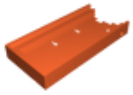
8x32-led-matrix-frame-diffuser_screen-split.stl



8x32-led-matrix-frame-diffuser_grid-split.stl



8x32-led-matrix-frame-backl-split.stl



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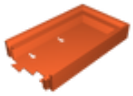


8x32-led-matrix-frame-backr-split.stl

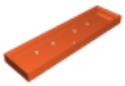
WHOLE - 3mf

4 files

8x32-led-matrix-frame-front-whole.3mf

8x32-led-matrix-frame-diffuser-whole.3mf

8x32-led-matrix-frame-back-whole.3mf



8x32-led-matrix-frame-back-no-sockets-whole.3mf

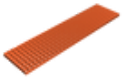
WHOLE - stl

6 files

8x32-led-matrix-frame-front-whole.stl

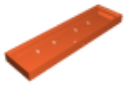
8x32-led-matrix-frame-diffuser-whole.stl

8x32-led-matrix-frame-diffuser_screen-whole.stl



8x32-led-matrix-frame-diffuser_grid-whole.stl

8x32-led-matrix-frame-back-whole.stl



8x32-led-matrix-frame-back-no-sockets-whole.stl

8x32-led-matrix-frame.step

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